Clapeyron and Clausius-Clapeyron equations

Consider a beaker of liquid phase in equilibrium with its gas phase. Because $dT = dP = 0$,

$$dG = -SdT + VdP + \mu_l dN_l + \mu_g dN_g = \mu_l dN_l + \mu_g dN_g. \tag{1}$$

Because the total number of molecules is conserved,

$$N_l + N_g = N = \text{constant}$$

$$\Rightarrow dN_l + dN_g = dN = 0 \Rightarrow dN_l = -dN_g. \tag{2}$$

Then Eq. (1) becomes

$$dG = (\mu_g - \mu_l)dN_g. \tag{3}$$

The condition for equilibrium at constant T and P is $dG = 0$, so

$$\mu_g(T, P) = \mu_l(T, P). \tag{4}$$

From Eq. (4), we have

$$d\mu_g(T, P) = d\mu_l(T, P). \tag{5}$$

Since μ_j is a function of (T, P), $d\mu$ can be written as

$$d\mu_j(T, P) = \left(\frac{\partial \mu}{\partial T}\right)_{P,N_j} dT + \left(\frac{\partial \mu}{\partial P}\right)_{T,N_j} dP. \tag{6}$$

With the Maxwell relations, we have

$$\left(\frac{\partial \mu_j}{\partial T}\right)_{P,N_j} = -\left(\frac{\partial S_j}{\partial N_j}\right)_{T,P} = -\bar{S}_j \tag{7a}$$

$$\left(\frac{\partial \mu_j}{\partial P}\right)_{T,N_j} = -\left(\frac{\partial V_j}{\partial N_j}\right)_{T,P} = \bar{V}_j \tag{7b}$$

where $\bar{S}_j$ and $\bar{V}_j$ are the molar entropy and volume of phase j, respectively. Substituting Eqs. (7a) and (7b) into Eq. (6) gives

$$d\mu_j(T, P) = -\bar{S}_j dT + \bar{V}_j dP. \tag{8}$$

Substituting Eq. (8) into Eq. (5) leads to

$$d\mu_g = -\bar{S}_g dT + \bar{V}_g dP = d\mu_l = -\bar{S}_l dT + \bar{V}_l dP \tag{9}$$

Rearranging Eq. (9) gives

$$\frac{dP}{dT} = \frac{\bar{S}_g - \bar{S}_l}{\bar{V}_g - \bar{V}_l} = \frac{\Delta\bar{S}}{\Delta\bar{V}}, \tag{10}$$

where $\bar{S} = \bar{S}_g - \bar{S}_l$ and $\bar{V} = \bar{V}_g - \bar{V}_l$. At the phase transition, $\Delta\mu = \Delta\bar{H} - T\Delta\bar{S}$, so

$$\Delta\bar{H} = T\Delta\bar{S}, \tag{11}$$

where $\Delta\bar{H} = \bar{H}_g - \bar{H}_l$. Substituting Eq. (11) into Eq. (10) yields the Clapeyron equation:

$$\frac{dP}{dT} = \frac{\Delta\bar{H}}{T\Delta\bar{V}}, \tag{12}$$

The molar volume of the gas phase is much larger than the molar volume of the liquid or solid phase, so $\Delta V = \bar{V}_g - \bar{V}_l \approx \bar{V}_g = RT/P$. Substituting this expression into Eq. (12) gives the Clausius-Clapeyron equation:

$$\frac{dP}{dT} = \frac{P\Delta\bar{H}}{RT^2},$$

Rearranging Eq. (13) gives

$$\frac{dP}{P} = \frac{\Delta\bar{H}}{RT^2}dT,$$

Assuming $\Delta\bar{H}$ is independent of temperature and integrating both sides of Eq. (13) gives

$$\begin{aligned} \int_{P_1}^{P_2} \frac{dP}{P} &= \int_{T_1}^{T_2} \frac{\Delta\bar{H}}{RT^2}dT, \\ \Rightarrow \ln\frac{P_2}{P_1} &= -\frac{\Delta\bar{H}}{R}\left(\frac{1}{T_2} - \frac{1}{T_1}\right). \end{aligned} \tag{13}$$